

Collateral Reuse and Financial Stability

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This presentation represents the views of the authors and should not be interpreted as reflecting the views of the Federal Reserve System or other members of its staff.

Motivation

- Securities financing transactions (SFTs): use securities to borrow cash or vice versa (repo, reverse repo, securities lending, and etc.)

As of March 2025:

- Outstanding SFTs in EU > **€15 trillion**
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 - Outstanding repos/reverse repos in the US > **\$6 trillion**
- Collateral posted by one side can be reused (rehypothecated) by the other side
 - Infante et al. (2020) find that the collateral multiplier (degree of collateral reuse) for U.S. Treasuries is around 10
- Policy makers have discussed potential systemic risks stemming from collateral reuse (Aitken and Singh, 2010; FSB, 2017)
- However, the industry has pushed back against this idea (Hill, 2014; ICMA 2025)

Research Question: What is the effect of collateral reuse on financial stability?

- Many papers on collateral reuse with leverage, liquidity, safe asset demand, risks of lender default, and collateral runs
 - BUT, all these factors confound the actual effect of collateral reuse

Research Question: What is the effect of collateral reuse on financial stability?

- Many papers on collateral reuse with leverage, liquidity, safe asset demand, risks of lender default, and collateral runs
 - BUT, all these factors confound the actual effect of collateral reuse
- **Isolated effects** of collateral reuse on financial stability are unclear:
 - Holding counterparty liabilities fixed, greater collateral reuse can guarantee and protect more debt obligations with fewer assets
 - More debt obligations depend on the same collateral, whose value can potentially drop
 - More collateral reuse can incentivize agents to take on greater risk
- Main contribution: new model to isolate effects of collateral reuse

Relation to the Literature 1/2

- Contagion in financial networks:
Allen and Gale (2000); Eisenberg and Noe (2001); Elliott, Golub, and Jackson (2014); Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015); Donaldson et al. (2022); **Chang and Chuan (2024)**
- Collateral reuse:
Infante (2019); Gottardi, Maurin, and Monnet (2019); Park and Kahn (2019); Infante and Vardoulakis (2021); Chang (2021); Luu et al. (2021); Maurin (2022); Brumm et al. (2023); Infante and Saravay (2024)
- Multiple equilibria in financial networks:
Rogers and Veraart (2013); Roukny, Battiston, and Stiglitz (2018); Bernard, Capponi, and Stiglitz (2022); Capponi, Corell, and Stiglitz (2022); Jackson and Pernoud (2024)

Relation to the Literature 2/2

- Equilibrium selection:
 - Global games:
Carlsson and Van Damme (1993); Morris and Shin (1998); Bernardo and Welch (2004); Goldstein and Pauzner (2005); Kuong (2021); Kashyap, Tsomocos, and Vardoulakis (2024); Eisenbach and Phelan (2025)
 - Best response dynamics:
Gilboa and Matsui (1991); Matsui (1992); Mäder (2024)
- Systemic risk and excessive risk-taking:
Castiglionesi, Feriozzi, and Lorenzoni (2019); Elliott, Georg, and Hazell (2021); Altinoglu and Stiglitz (2023); Jackson and Pernoud (2024); Shu (2024)

Preview of Main Results

1. There are three equilibria:
 - (1) best (maximum) equilibrium
 - (2) intermediate equilibrium
 - (3) worst (minimum) equilibrium

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4. When agents' risk-taking decisions are endogenous, social surplus in the worst eqm decreases in collateral reuse

Collateral reuse $\uparrow \Rightarrow$ Crisis is **less likely** but **more severe**

Implications

- Collateral reuse alone is not a concern for financial stability
- Collateral reuse can still negatively impact financial stability through indirect effects on risk-taking decisions
- Supporting the price and liquidity of collateral assets can significantly improve social surplus (dealer of last resort, SRF, and etc.)

Model of Contagion

Multiple Equilibria and Contagion

Illustrative Example

Collateral Reuse and Changes in Equilibria

Equilibrium Selection

Global Games

Best Response Dynamics

Endogenous Risk Taking

Conclusion

Appendix

Model Overview

- Based on Chang and Chuan (2024), which extends Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015)
- Key features:
 1. **Assets** can be used as collateral of a debt contract
 2. Asset price is endogenously determined
- Main objects of interest: network structure and collateral amount

Agents, Goods, and Long-term Project

- Three periods: $t = 0, 1, 2$
- Two goods in economy: cash (e) and an asset (h)
 - Cash is the only consumption good and is storable
 - Asset yields s amount of cash at $t = 2$

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 - Long-term investment project gives ξ amount of cash at $t = 2$
 - If liquidated l_j prematurely, receive ζl_j at $t = 1$ (assume $\zeta \rightarrow 0$)
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- State of the liquidity shocks: $\omega \equiv (\omega_1, \omega_2, \dots, \omega_n) \in \Omega$
 $\implies$ Liquidity shock ϵ = **senior liability**

Collateralized Debt Contracts and Constraints

- Debt contract btw lender i and borrower j from $t = 0$ to $t = 1$
- Debt amount d_{ij} and collateral(-to-debt) ratio c_{ij}
- Lender has full recourse and collateral transfer is frictionless

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- Debt contract btw lender i and borrower j from $t = 0$ to $t = 1$
- Debt amount d_{ij} and collateral(-to-debt) ratio c_{ij}
- Lender has full recourse and collateral transfer is frictionless
- Collateral constraints (allow reuse of collateral):

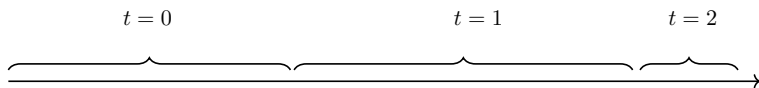
$$\sum_{k \in N} c_{jk} d_{jk} + h_j \geq \sum_{i \in N} c_{ij} d_{ij} \quad \forall j \in N$$

- Resource constraints:

$$\sum_{i \in N} h_i \geq \sum_{i \in N} c_{ij} d_{ij} \quad \forall j \in N$$

- Collateralized debt network: $(C, D) \equiv ([c_{ij}]_{i,j \in N}, [d_{ij}]_{i,j \in N})$

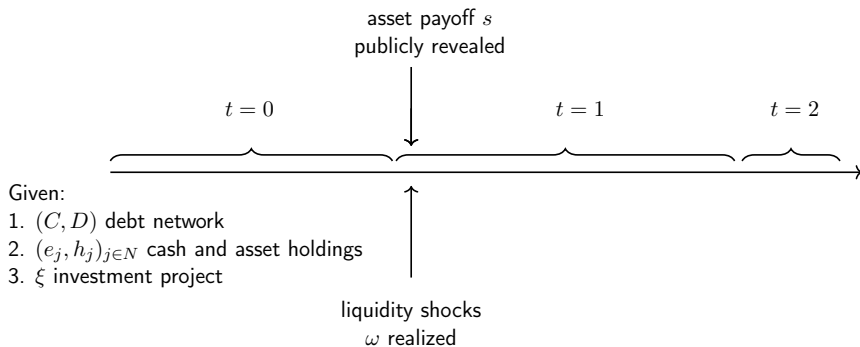
Timeline



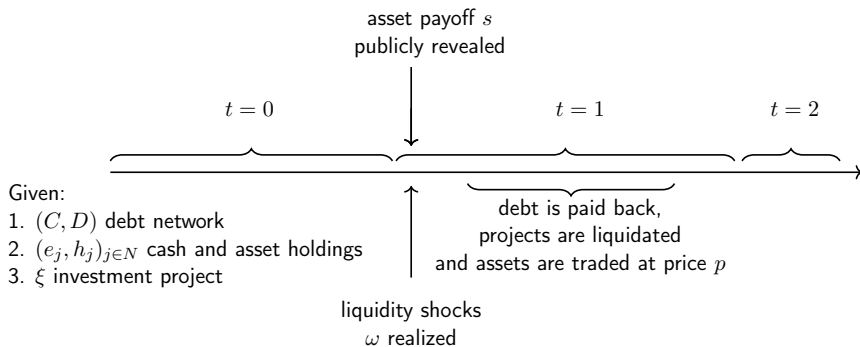
Given:

1. (C, D) debt network
2. $(e_j, h_j)_{j \in N}$ cash and asset holdings
3. ξ investment project

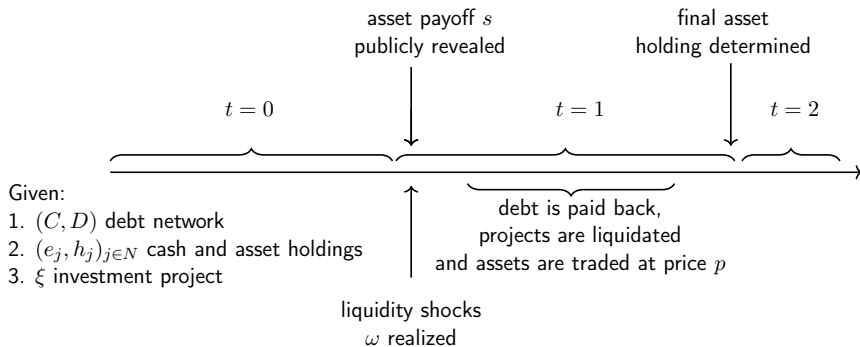
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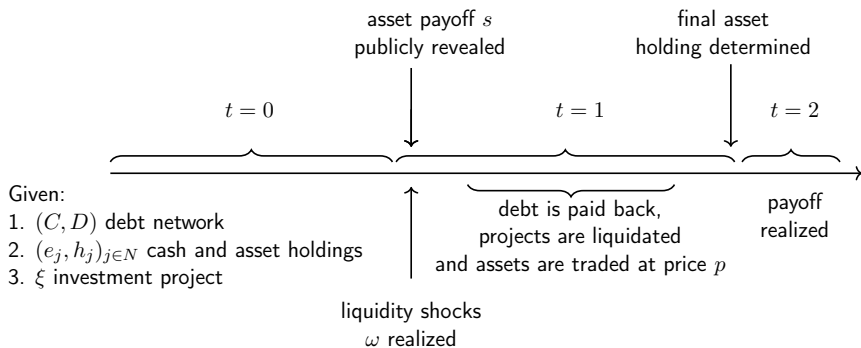
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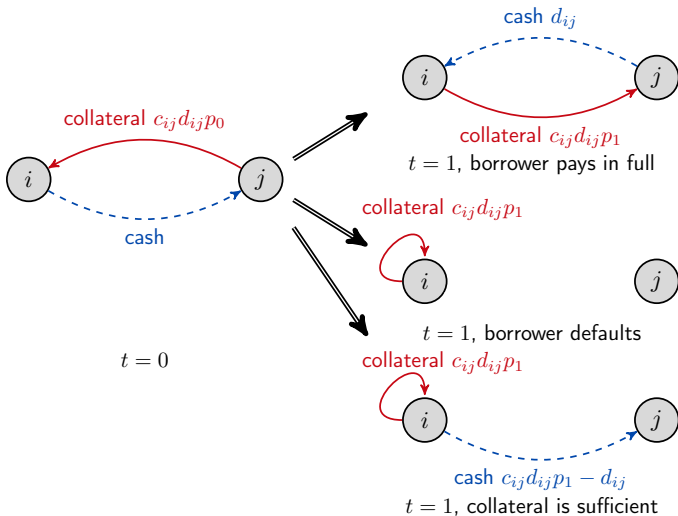
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Flows of Cash and Collateral



Payment Rule

- j 's total cash inflow:

$$a_j(p) \equiv e_j + h_j p + \sum_{k \in N} c_{jk} d_{jk} p - \sum_{i \in N} c_{ij} d_{ij} p + \sum_{k \in N} \underbrace{x_{jk}(p)}_{\text{actual payment to } j \text{ from } k}$$

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- j 's total cash outflow:

$$b_j(p, \omega) \equiv \sum_{i \in N} \underbrace{(d_{ij} - c_{ij} d_{ij} p)}_{\text{net debt amount}} + \omega_j \epsilon,$$

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- Net payment to agent i from agent j at asset price p at $t = 1$:

$$x_{ij}(p) = \min \left\{ d_{ij} - c_{ij} d_{ij} p, \quad q_{ij}(p) \left[a_j(p) + \sum_{i \in N} [c_{ij} d_{ij} p - d_{ij}]^+ - \omega_j \epsilon \right]^+ \right\}$$

$$\text{pro rata weights: } q_{ij}(p) = \frac{[d_{ij} - c_{ij} d_{ij} p]^+}{\sum_{k \in N} [d_{kj} - c_{kj} d_{kj} p]^+}$$

Agent Wealth

- Net wealth of agent j :

$$\begin{aligned} m_j(p) &\equiv a_j(p) - b_j(p) \\ &= e_j + h_j p + \sum_{k \in N} c_{jk} d_{jk} p - \omega_j \epsilon - \sum_{i \in N} d_{ij} + \sum_{k \in N} x_{jk}(p) \end{aligned}$$

- If $m_j(p) < 0$, then agent j defaults, i.e., $j \in \mathcal{D}(p)$
- Excess net wealth can be used to purchase assets

Fire-Sales

- Assume that agents sell up to its asset holdings amount
- *Fire-sale* amount of agent j :

$$\phi_j(p) = \min \{ [h_j p - m_j(p)]^+, h_j p \}$$

Agent's optimization problem

- If agent j 's net cash flow, $m_j(p) - h_j p$, covers payments
 $\implies \phi_j(p) = 0$
- If agent j 's net cash flow + sale of assets covers payments
 $\implies 0 < \phi_j(p) < h_j p$
- If agent j 's net cash flow + sale of assets does not cover payments
 $\implies \phi_j(p) = h_j p$

Market Clearing Condition

- Asset's fundamental value s is common knowledge, and $p \leq s$.
- Liquidity constrained price $p < s$, i.e., cash-in-the-market pricing
- Price of collateral (p) is determined by the mkt clearing cdn:

$$\sum_{j \in N} [m_j(p) - h_j p]^+ = \sum_{i \in N} \phi_i(p) \quad \text{if } 0 \leq p < s$$

$$\sum_{j \in N} [m_j(s) - h_j s]^+ \geq \sum_{i \in N} \phi_i(s) \quad \text{iff } p = s,$$

- Excess net wealth vs fire sale amount

Market Clearing Implications

- **Lemma.** If aggregate positive net wealth $\sum_{j \in N} [m_j(p)]^+ > 0$, then it is strictly increasing in the asset price p .

- **Lemma.** The market clearing asset price can be represented as

$$p = \min \left\{ \frac{\sum_{j \in N} [m_j(p)]^+}{\sum_{j \in N} h_j}, s \right\}$$

Full Equilibrium

Definition

*For given $(N, C, D, e, h, s, \omega)$, if payments $\{x_{ij}(p)\}$ satisfy the payment rule, $\{m_j(p)\}$ is determined by net wealth equation, the fire-sale amount $\{\phi_j(p)\}$ is determined by fire-sale equation, and price p clears the market, then $(\{x_{ij}\}, \{m_j\}, \{\phi_j\}, p)$ is a **full equilibrium**.*

Proposition (Chang and Chuan (2024))

For any given collateralized debt network, cash and asset holdings, asset payoff, and realization of shocks $(N, C, D, e, h, s, \omega)$, a full equilibrium always exists and is generically unique for a given equilibrium price. Furthermore, there exists a full equilibrium with the highest price among the set of full equilibria.

Contagion and Social Surplus

- For any given collateralized debt network and full equilibrium, the utilitarian social surplus in the economy at $t = 2$ is:

$$U = \sum_{i \in N} (\pi_i + T_i),$$

where $T_i \leq \epsilon$ is the transfer from agent i to senior creditors (liquidity shock) and π_i is agent i 's profit at $t = 2$

Lemma

For any full equilibrium, the social surplus in the economy is equal to

$$U = \sum_{j \in N} (e_j + h_j s + \xi) - \sum_{i \in N} l_i$$

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Regularity Assumptions

- Regular networks: $\sum_{i \in N} d_{ij} = \sum_{i \in N} d_{ji} = d$ for all $j \in N$
- Homogeneous endowments: $e_i = e_0$ and $h_i = h_0$ for all $i \in N$
- Uniform collateral ratio: $c_{ij} = c$ for all $i, j \in N$
 - Collateral ratio c is equivalent to the *degree of collateral reuse*
- Liquidity shock: $\epsilon > ne_0$ received by one agent

Collateral Ratio and Connectedness Thresholds

Proposition (Chang and Chuan (2024))

1. If $c \geq \bar{c}(s, n) \equiv \frac{1}{s}$, then no agent defaults in the maximum equilibrium for any given network D .
2. If $c \geq \underline{c}(s, n) \equiv \frac{d - (n - 1)e_0 + h_0s}{ds}$, then the asset price is $p = s$ in the maximum equilibrium for any given network D .

- Focus on cases in which collateral can play its role in mitigating contagion:

$$\underline{c}(s, n) \leq c$$

- Assume that agents are *sufficiently interconnected*

Details

Main Result 1: Three Unique Equilibria

Proposition

For any network D and any $c \geq \underline{c}(s, n)$, three different equilibria exist generically:

- 1. Maximum equilibrium: $p = s$ with the least number of defaults*
- 2. Minimum equilibrium: $p = 0$ with all agents default*
- 3. Intermediate equilibrium: $0 < p < s$ s.t. market clearing condition implies cash-in-the-market pricing, i.e., $p = \frac{\sum_{j \in N} [m_j(p)]^+}{nh_0}$*

- Significant swings in social surplus based on coordination of eqm
- Interventions can prevent bad equilibrium (dealer of last resort)

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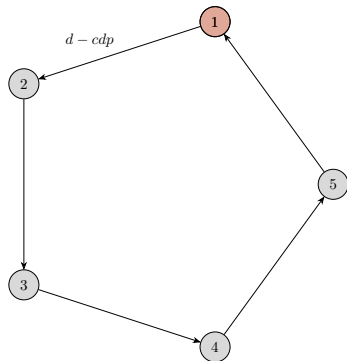
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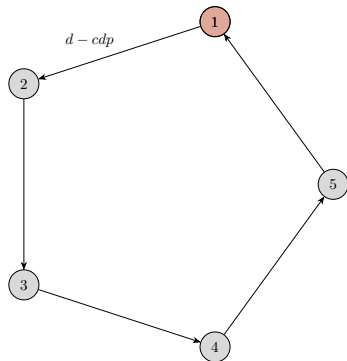
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Example: Maximum Equilibrium



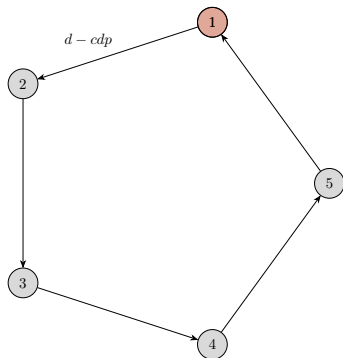
- Each agent has
 $e_0 = 2$: 2 units of cash
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 $d = 10$: debt amount
 $c = 1$: collateral ratio
- Asset's fair value $s = 1$
- Agent 1 hit by liquidity shock

Example: Maximum Equilibrium



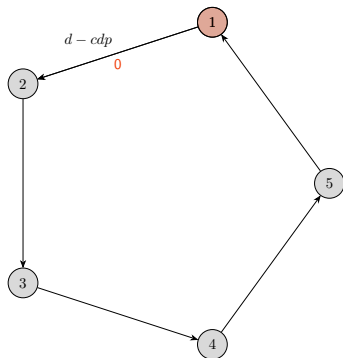
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- Agent 1 hit by liquidity shock
- If $p = 1$, net debt amount is $d - cdp = 10 - 1 \times 10 \times 1 = 0$, and no defaults
- Surviving agents' aggregate cash 8 can clear the 2 assets on sale at its fundamental value $p = s = 1$

Example: Minimum Equilibrium



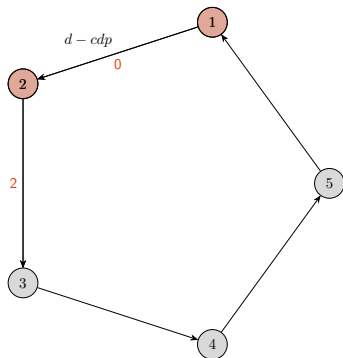
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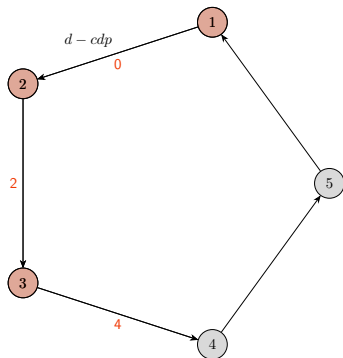
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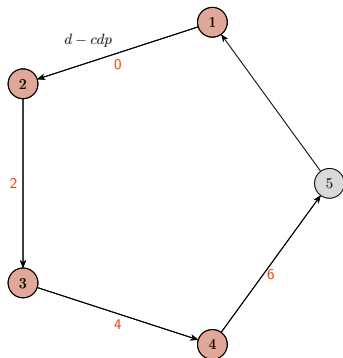
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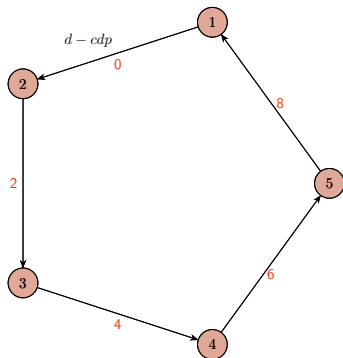
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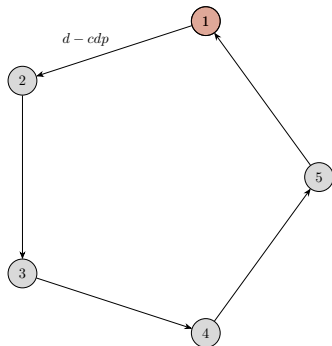
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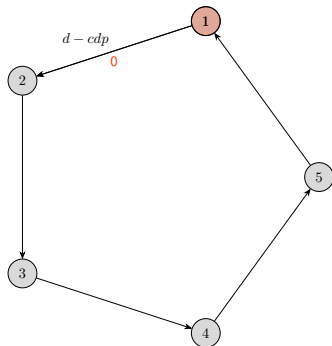
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- All agents default and liquidate their long-term projects, and $p = 0$

Example: Intermediate Equilibrium



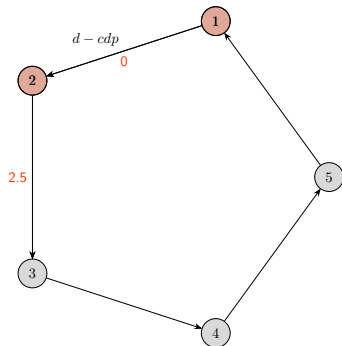
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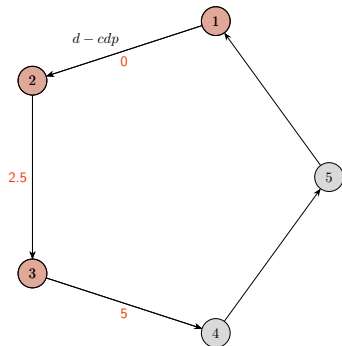
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- If $p = 0.25$, net debt amount is $d - cdp = 10 - 1 \times 10 \times 0.25 = 7.5$, and agent 2's remaining wealth is $2 + 2 \times 0.25 = 2.5$

Example: Intermediate Equilibrium



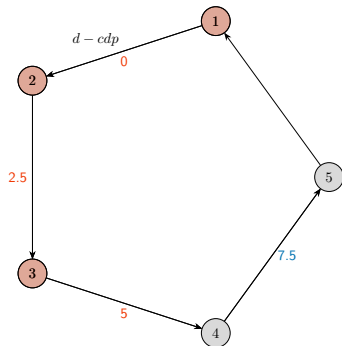
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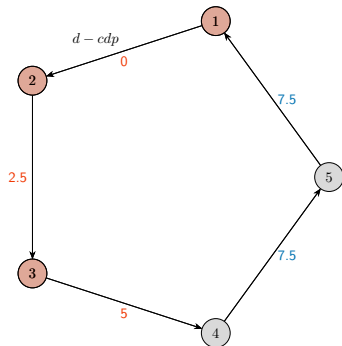
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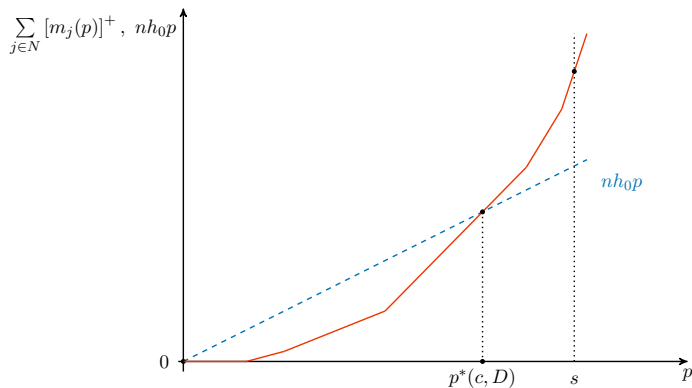
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- Agent 5 has 2 cash to buy 8 assets at $p = 2/8 = 0.25$

Illustration of Multiple Equilibria



$$p = \min \left\{ \frac{\sum_{j \in N} [m_j(p)]^+}{nh_0}, s \right\}$$

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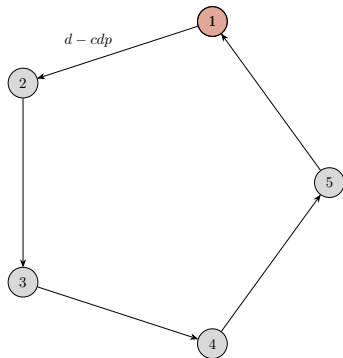
Main Result 2: Collateral Reuse Decreases Intermediate Equilibrium Price

Proposition

The intermediate equilibrium price $p^(c, D)$ is decreasing in the degree of collateral reuse c regardless of the network structure D .*

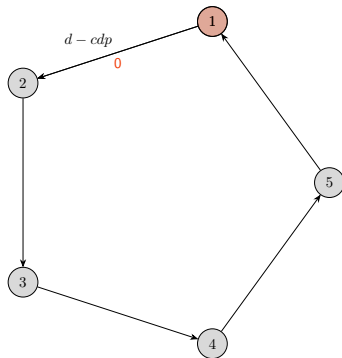
- $cdp \uparrow$ as $c \uparrow$, so $p \downarrow$ to equate the market clearing condition
- Aggregate net wealth becomes more sensitive to collateral price p
- Overall effect can lead to more defaults, as the aggregate net wealth in the intermediate equilibrium decreases

Intermediate Equilibrium Example: $c \downarrow \implies p \uparrow$



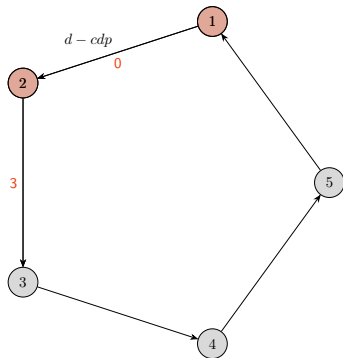
- Each agent has
 - $e_0 = 2$: 2 units of cash
 - $h_0 = 2$: 2 units of assets
 - $d = 10$: debt amount
 - $c = 0.6$: collateral ratio
- Asset's fair value $s = 1$
- Agent 1 hit by liquidity shock

Intermediate Equilibrium Example: $c \downarrow \implies p \uparrow$



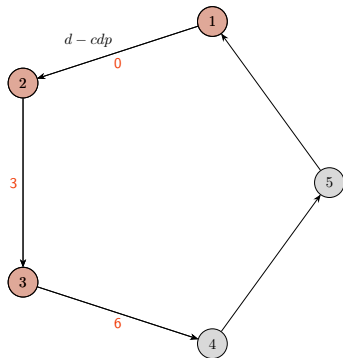
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- If $p = 0.5$, net debt amount is $d - cdp = 10 - 0.6 \times 10 \times 0.5 = 7$, and agent 2's remaining wealth is $2 + 2 \times 0.5 = 3$

Intermediate Equilibrium Example: $c \downarrow \implies p \uparrow$



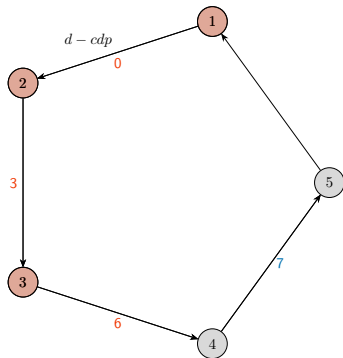
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Intermediate Equilibrium Example: $c \downarrow \implies p \uparrow$



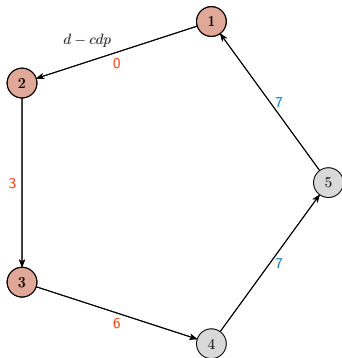
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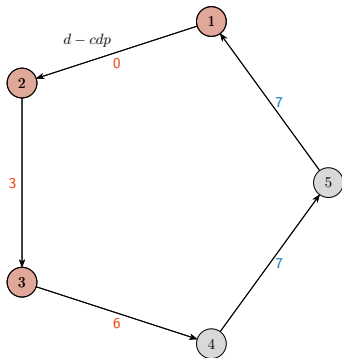
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Intermediate Equilibrium Example: $c \downarrow \implies p \uparrow$



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- Agent 4 and 5 have 1 and 2 cash, respectively, to buy 6 assets at $p = 3/6 = 0.5$

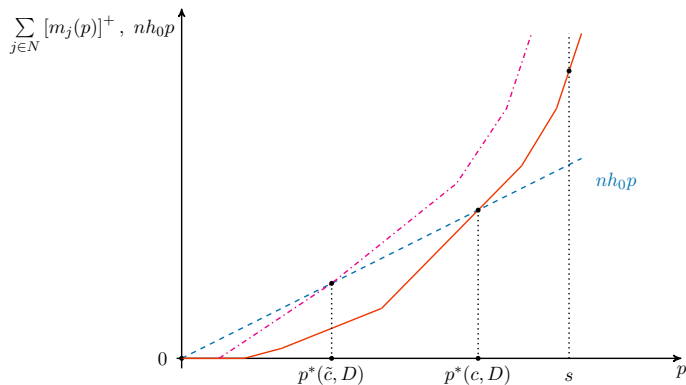
Intermediate Equilibrium Example: $c \downarrow \implies p \uparrow$



- lower c ($1 \rightarrow 0.6$)
but higher p ($0.25 \rightarrow 0.5$)

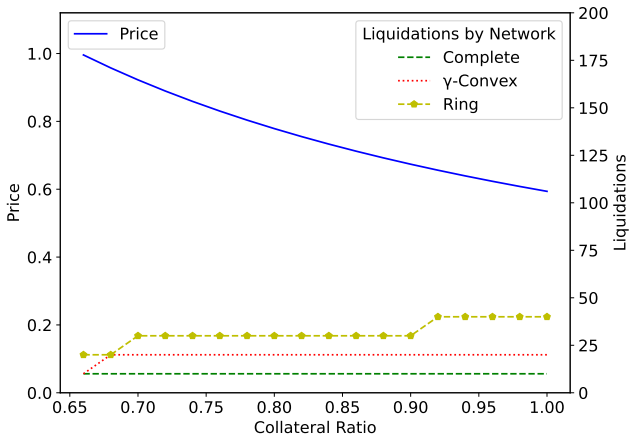
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Illustration of Collateral Reuse and Changes in Equilibria



$$p = \min \left\{ \frac{\sum_{j \in N} [m_j(p)]^+}{nh_0}, s \right\}$$

Numerical Examples of Intermediate Equilibrium



- Even though the amount of liquidations can differ, the intermediate equilibrium price is the same across three networks

Model of Contagion

Multiple Equilibria and Contagion

Illustrative Example

Collateral Reuse and Changes in Equilibria

Equilibrium Selection

Global Games

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Equilibrium Selection

- Extend our model to incorporate equilibrium selection:
which equilibrium out of the three will be realized?
- We consider both global games and best response dynamics
 - Results hold for both models of equilibrium selection
- Intermediate eqm price plays an important role in eqm selection

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Equilibrium Selection and Global Games

- Building on top of the Global Games literature: Carlsson and Van Damme (1993), Morris and Shin (1998), Bernardo and Welch (2004), Goldstein and Pauzner (2005), Infante and Vardoulakis (2021), **Kuong (2021)**, Kashyap et al. (2024), Eisenbach and Phelan (2025)
- Strategic substitutability due to market forces
(one agent's sales can be another agent's buying opportunity)
- Strategic complementarity arises due to payment obligations and contagion (cf. sequential servicing constraint, moral hazard and self-fulfilling fire sales)

Global Games Model

- Value of the asset payoff $\theta \sim U[0, \bar{\theta}]$, where $\bar{\theta} \geq s$
- Each agent receives a noisy signal $\theta_i = \theta + \psi_i$, $\psi_i \sim_{iid} U[-\psi, \psi]$
- Assume each individual agent is distributed on a continuum (a continuum of identical $i \in N$ with a mass of one)

Details

Main Result 3: Equilibrium Selection GG

Proposition

In the global games setup with $\psi \rightarrow 0$, the following hold:

- 1. For any $\theta \geq p^*(c, D)$, the equilibrium price is $p = \theta$.*
 - 2. For any $\theta < p^*(c, D)$, the equilibrium price is $p = 0$, and all agents default.*
 - 3. The likelihood of the minimum equilibrium with $p = 0$ decreases as the degree of collateral reuse c increases.*
-
- The unique intermediate equilibrium price $p^*(c, D)$ is the threshold for equilibrium selection
 - Any price above the threshold can satisfy the market clearing condition with $\sum_{j \in N} [m_j(\theta)]^+ \geq nh_0\theta$, for any $\theta \geq p^*(c, D)$
 - Any price below the threshold leads to the minimum equilibrium as $\sum_{j \in N} [m_j(\theta)]^+ < nh_0\theta$ for any $0 < \theta < p^*(c, D)$

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Best Response Dynamics

- Alternative equilibrium selection model based on Gilboa and Matsui (1991), Matsui (1992), and Mäder (2024)
- BRD: Each agent responds with a best response to all other agents' previous actions in each round
⇒ a best response path can converge to a Nash equilibrium
- Applicability: After observing a sudden price decline, agents who initially decided not to participate in the market, can and optimally choose to participate in the market
- Microfoundation of Walrasian tâtonnement

Details

Main Result 3': Equilibrium Selection BRD

Proposition

In the BRD-RB setup, the likelihood of realization of the minimum equilibrium is proportional to $p^(c, D)$. Moreover, the likelihood of the minimum equilibrium decreases as the degree of collateral reuse c increases.*

- **Basin of attraction** of an eqm is the set of all initial conditions giving rise to at least one BRD path ending in that eqm
- Basin of attraction of the maximum equilibrium is $(p^*(c, d), s]$
- As the degree of collateral reuse increases, the financial system is less likely to be perturbed by random noise and end up in $p = 0$

Illustration of Equilibrium Selection

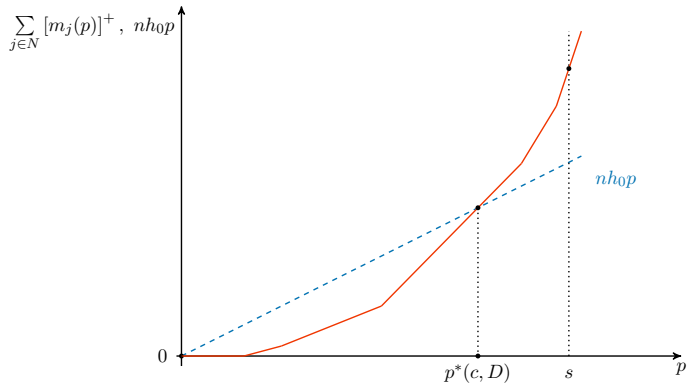
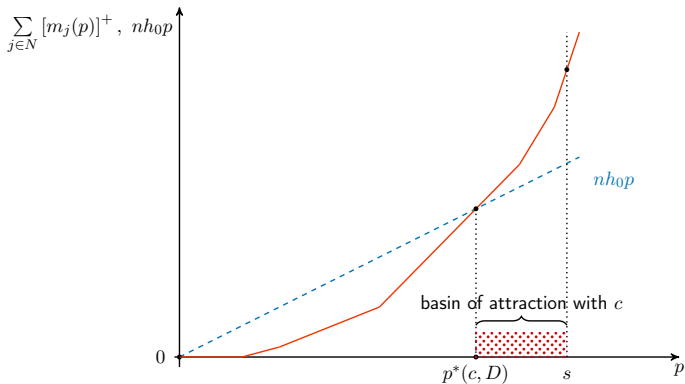
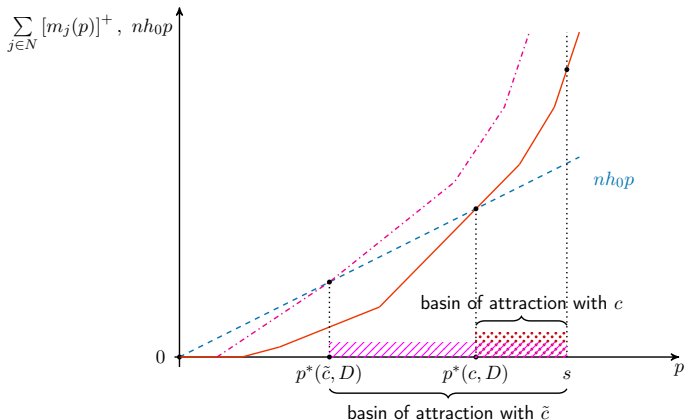


Illustration of Equilibrium Selection



- Any starting price within the basin of attraction will end up at $p = s$

Illustration of Equilibrium Selection



- Any starting price within the basin of attraction will end up at $p = s$

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Collateral Reuse and Other Factors

- Degree of collateral reuse is often correlated with other factors: leverage, collateral circulation and scarcity, collateral runs and lender default, and length of lending chain (Gottardi et al., 2019; Infante, 2019; Park and Kahn, 2019; Infante and Vardoulakis, 2021; Chang, 2021; Maurin, 2022; Brumm et al., 2023; Infante and Saravay, 2024)
- We extend our model further to incorporate agents' endogenous risk taking behavior at $t = 0$
- All other factors can amplify our mechanism
- For simplicity, use BRD-RB equilibrium selection model (with uniformly distributed RB)

Risk-Taking Tradeoff

- Each agent has a cash endowment e_{-1} at the beginning of $t = 0$
- Agents allocate cash into
 - cash holdings e_0
 - investment in the long-term project ξ

Risk-Taking Tradeoff

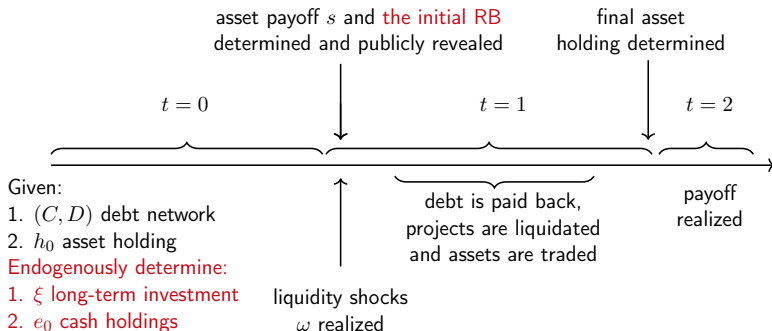
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- Long-term project investment
 - Each unit of cash invested gives $R > 1$ at $t = 2$
 - If liquidated, the agent suffers *liquidation cost* $K(\xi)$
 $K(\xi)$ is convexly increasing ($dK/d\xi > 0$ and $d^2K/d\xi^2 > 0$)

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 $K(\xi)$ is convexly increasing ($dK/d\xi > 0$ and $d^2K/d\xi^2 > 0$)
- Expected cost of liquidation:

$$p^*(c, D)K(\xi) + (s - p^*(c, D))\frac{1}{n}K(\xi)$$

Timeline with Endogenous Risk Taking



Agent's Optimization Problem at Date 0

$$\begin{aligned} \max_{e_0, \xi} & -p^*(c, D)K(\xi) + (s - p^*(c, D)) \left[\frac{n-1}{n} (\xi + \tilde{\pi}_j(e_0, D)) - \frac{1}{n} K(\xi) \right] \\ \text{s.t.} & \quad e_0 + \xi/R \leq e_{-1} \end{aligned}$$

where $\tilde{\pi}_j(e_0, D) \equiv e_0 + h_0 s - (d - cds) + E_j \left[\sum_{j \neq i} x_{ji}(s) | \omega_j = 0 \right]$ is the expected profit of agent j excluding the long-term investment payoff, when j is not shocked

- FOC:

$$\underbrace{\left[p^*(c, D) + (s - p^*(c, D)) \frac{1}{n} \right]}_{\substack{\text{increasing in } p^*(c, D), \\ \text{hence decreasing in } c}} K'(\xi) = \underbrace{(s - p^*(c, D))}_{\substack{\text{decreasing in } p^*(c, D), \\ \text{hence increasing in } c}} \frac{n-1}{n} \left(1 - \frac{1}{R} \right)$$

- As c increases, agent j is less worried about liquidation cost
- Optimal investment amount ξ^* is increasing in c

Main Result 4: Fewer But More Severe Crises

Corollary (Empirical Prediction)

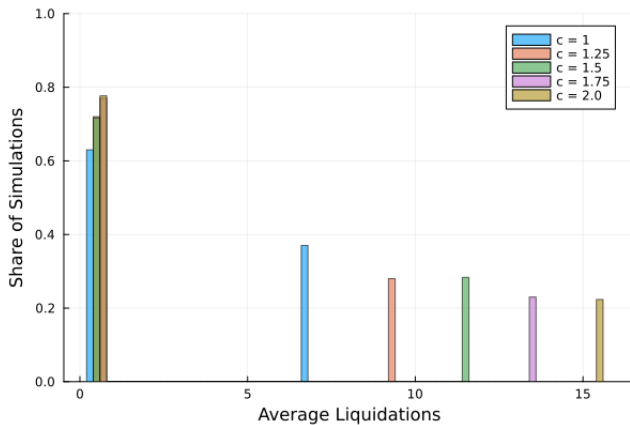
Holding all else equal, a higher degree of collateral reuse leads to a lower likelihood of crisis but a greater severity of crisis (lower social surplus) when it occurs.

- This result aligns well with the recent literature on excessive risk-taking behavior by individuals and systemic risk (Castiglionesi et al., 2019; Elliott et al., 2021; Galeotti and Ghiglino, 2021; Altinoglu and Stiglitz, 2023; Jackson and Pernoud, 2024; Shu, 2024)
- Our results demonstrate how endogenous risk-taking behavior can be exacerbated by an increase in collateral reuse

Results are Robust to Various Extensions

- Non-trivial liquidations ($\zeta > 0$)
- Multiple agents hit by liquidity shocks
- Size heterogeneity (scaling each agent's balance sheet differently)
- Non-regular network structures (e.g., core-periphery networks)
- External traders and non-zero minimum equilibrium price

Random Networks Simulations



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Conclusion

- A model with both debt and collateral market contagion and endogenous fire-sale prices with multiple equilibria
- Even with the same degree of collateral reuse, there can be extreme swings in social surplus across equilibria
- Collateral reuse alone is not a concern for financial stability (lower likelihood of crises)
- Degree of collateral reuse can still negatively impact financial stability through its effects on risk-taking choices (lower likelihood but greater severity of crises)
- Supporting price and liquidity of collateral assets can be a critical policy intervention

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Agent's Optimization Problem at $t = 1$

- Agent j maximizes long-term profit π_j at $t = 2$
- Decides: cash (e) and assets (h)
- Decisions subject to wealth, liabilities, and belief θ_j , which is s for now

$$\begin{aligned} \max_{e,h} \pi_j &= e + hs + \xi \mathbb{1} \{a_j(p) > b_j(p)\} \\ \text{s.t.} \quad e + hp &= [a_j(p) - b_j(p)]^+ \\ e &\geq 0, \quad h \geq 0 \end{aligned}$$

- Agent j will buy assets using all the available budget if $p < s$

Harmonic Distance 1/2

Definition

The harmonic distance from agent i to agent j is

$$\mu_{ij} = 1 + \sum_{k \neq j} \left(\frac{d_{ik}}{d} \right) \mu_{kj},$$

with the convention that $\mu_{ii} = 0$ for all i .

- We focus on the set of networks such that any network has harmonic distances smaller than $\mu^*(0)$ when $p = 0$

Harmonic Distance 2/2

Proposition (Chang and Chuan (2024))

Suppose that agent j is under a negative liquidity shock of $\epsilon > ne_0$. Then, there exists $\mu^(p) = (d - cdp)/(e_0 + h_0p)$, and the following holds:*

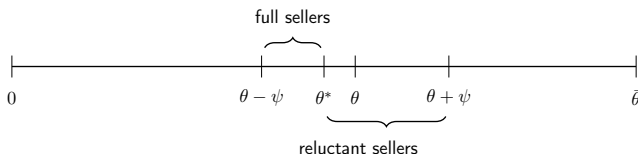
- 1. If there is a nonempty set S such that agent $i \in S$ does not default, then the equilibrium price is either $p = s$ or determined by*

$$\mathbb{1}'G\mu_{sj} = \frac{d - cdp}{e_0 + h_0p}\mathbb{1}'G\mathbb{1} + \frac{nh_0p}{e_0 + h_0p},$$

where μ_{sj} is the vector of harmonic distances from agents in S to j , G is a $|S| \times |S|$ non-singular M-matrix, and $\mathbb{1}$ is a vector of ones. Furthermore, if $\mu_{ij} < \mu^(p)$, then agent i defaults.*

- 2. If all agents default, then the equilibrium price is $p = 0$ and $\mu_{ij} < \mu^*(0)$ for all i .*
- 3. If $\mu^*(p) < 1$ for the equilibrium price p , then no other agents default.*

Global Games Supply and Demand



- Marginal buyer $\theta^* \in [\theta - \psi, \theta + \psi]$
- Full sellers: $\frac{\theta^* - (\theta - \psi)}{2\psi}$ vs. Reluctant sellers: $\frac{\theta + \psi - \theta^*}{2\psi}$
- Endogenous fire-sale amount of j in the entire continuum:

$$\phi_j(p) = \min \left\{ \frac{\theta + \psi - \theta^*}{2\psi} [h_j p - m_j(p)]^+ + \frac{\theta^* - (\theta - \psi)}{2\psi} h_j p, h_j p \right\}$$

- Endogenous demand:

$$\sum_{j \in N} \frac{\theta + \psi - \theta^*}{2\psi} [m_j(p) - h_j p]^+$$

Intermediate Game of BRD

- Agent i in N maximizes its profits π_i by choosing an action $\alpha_i \in A = [0, h_0 p]$

- Strategic decision changes the fire sales amount:

$$\phi_i(p) = \min \{ [h_0 p - m_i(p)]^+ + \alpha_i, h_0 p \}$$

- Agents continuously revise their actions for each $\tau \in [0, 1]$ (internal timing for the BRD path)
- A BRD path is a continuous and right-differentiable function $\alpha : [0, 1] \mapsto \mathcal{A}$ such that $\alpha'(\tau) \in \delta [\beta(\alpha(\tau)) - \alpha(\tau)]$ for some $\delta > 0$, for each $\tau \in [0, 1)$, and for any $\alpha(0) \in \Pi_{i \in N} A$
- Agents start with random beliefs (RB) $\kappa : \Theta \mapsto \Pi_{i \in N} \Pi_{j \neq i} A$